

Attested Transition Execution for DAZE-Style Simulator Backdoors

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Abstract

Reinforcement-learning (RL) agents are increasingly trained on third-party simulators, making the simulator a supply-chain trust anchor. DAZE shows that a malicious simulator wrapper can implant reward-free backdoors by changing what transition the learner observes after it submits an action without changing the action itself, so action-level logging cannot detect it [1]. Prior RL backdoor work mainly studies policy/agent triggers, competitive backdoors, or sparse targeted attacks [2, 3, 4]; we instead protect the simulator data plane. We study the complementary defense problem under an explicit trust boundary: the simulator core, configuration/assets, random-number state, observation and termination code, reward inputs, and actuator transform are measured and trusted, while the wrapper is untrusted. *Attested Transition Execution* (ATE) accepts a learner-visible update only when an approved measured step relation, exact replay, or an enrolled receipt service vouches for the full transition record. The headline result is zero admitted tampered updates: on original DAZE Hopper-v5 and Reacher-v5 PPO, attack ASR reaches at least 0.8435 and 0.9843, while ATE restores clean-level ASR and matched benign-attested return with ASR-lift 0 and utility 1.0000 over 3.94M defended receipt checks; on SAC Reacher-v5, ASR drops from 0.9996 to 0 with zero forged updates.

Claim and Contributions

Claim. For an approved measured simulator closure with an untrusted wrapper, DAZE-style transition tampering cannot feed forged learner-visible updates into training without failing ATE admission. Positive results support this claim across PPO/SAC continuous-control MuJoCo, Gymnasium classic control, and a custom discrete DQN replay simulator.

Contributions. (i) We frame simulator-wrapper backdoor defense as transition attestation rather than post-hoc anomaly detection. (ii) We show action-only logging is insufficient: a Hopper termination-tamper stress case preserves $a_{\text{sub}} = a_{\text{exec}}$ yet reaches ASR 0.982. (iii) ATE restores original-DAZE Hopper/Reacher to clean-level ASR/return, admits zero poisoned updates over 3.94M defended checks, and reduces the action-equality Hopper termination-tamper stress case to ASR 0.009.

Design

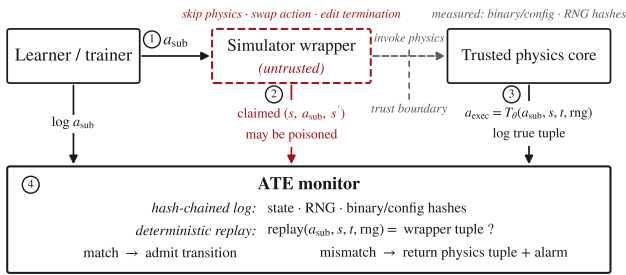


Figure 1. ATE places the admission gate at the transition boundary, not at wrapper self-report. The monitor admits a learner update only when a trusted step relation or receipt path vouches for the submitted action, state/RNG, and learner-visible fields. This makes action-only logging insufficient for wrapper-level DAZE tampering.

ATE accepts a transition only when the record

$$(s, o, a, T_{\theta}(a, s, t, \text{rng}), s', o', r, d, \tau)$$

matches the approved step relation. T_{θ} is a measured declarative actuator transform for legitimate clipping, scaling, action repeat, torque saturation, or safety projection. Exact mode replays deterministic CPU simulators. Receipt mode follows measured-evidence principles from remote attestation architectures [5]: it binds the public key to code, dependencies, closure hash, verifier policy, socket identity, nonce, service id, and key epoch; admission checks signatures, sequence numbers, nonces, lane identity, and hash-chain continuity.

Positive Learning Evidence

Setting	Learner	Clean	Attack	ATE	Utility	Admit
Hopper-v5	PPO	0.0177	0.8435	0.0177	1.0000	0
DAZE Reacher-v5	PPO	0.0000	0.9843	0.0000	1.0000	0
DAZE Reacher-v5	SAC	0.0000	0.9996	0.0000	clean-level	0
DAZE CartPole record	DQN	0.415	0.961	0.415	clean-level	0/80,350
tamper						
Discrete replay	DQN	0.000	0.908/0.933	0.000	0.941	0/all
relabel						

Table 1. Positive learning evidence. Clean, Attack, and ATE report attack success rate (ASR); rows are promoted only when the attack attaches and ATE admits zero forged updates while preserving utility. The SAC Reacher row adds cross-learner evidence, while the DQN rows show that the gate is not only a MuJoCo/PPO artifact.

The PPO rows are five-seed original-DAZE promotion runs; Attack is minimum unfiltered ASR and ATE/Clean are means. ATE performs 1.49M Hopper and 2.45M Reacher defended checks, admits zero tamper, and matches benign return (1467.62; -6.47). CartPole’s natural target base rate makes clean-level ASR the correct endpoint.

Systems Evidence

Audit or benchmark	Rows/records	Failures
MuJoCo promotion validation	168	0
Software-root protocol v3	57	0
AMC-ATE v2 stress	71	0
Brax/JAX GPU receipts	1.57M	0

Table 2. Systems evidence for the admission path. Rows/records count validation checks or receipt records; failures count accepted violations or false rejects in the corresponding audit. These results test the software-root protocol, the MuJoCo promotion gate, stress cases, and batched GPU receipt handling.

The v3 protocol audit rejects rogue keys, closure/policy drift, stale nonces, key swaps, rollback, transition edits, stale/wrong-key receipts, lane mismatch, stale anchors, preflight violations, and transform-governance attacks. Batch receipts amortize cost: CartPole costs 11.18 μs /record at $1.20\times$ with 7.0 bytes/record; MuJoCo overhead is $1.01\text{--}1.23\times$; Brax/JAX GPU receipts cost $1.30\text{--}1.64\times$ with 0.137 bytes/record.

Boundary and Discussion Prompt

The positive claim is intentionally scoped: the approved closure includes the simulator core, configuration/assets, RNG schedule, observation and termination code, reward inputs, and allowed actuator transforms. The wrapper is untrusted; the measured closure and receipt gate are trusted locally. Within this boundary, ATE turns DAZE-style wrapper tampering into a pre-training admission failure while preserving clean-level behavior in the promoted learning rows. **Prompt:** what root should bind receipt keys for remote batched RL-TPM/TEE quotes, signed step services, or scheduler-enforced measured jobs?

References

- [1] E. Rathbun, W. W. Lin, A. Oprea, and C. Amato. Beware Untrusted Simulators – Reward-Free Backdoor Attacks in Reinforcement Learning. ICLR, 2026.
- [2] P. Kiourti, K. Wardega, S. Jha, and W. Li. TrojDRL: Trojan Attacks on Deep Reinforcement Learning Agents. DAC, 2020.
- [3] L. Wang, Z. Javed, X. Wu, W. Guo, X. Xing, and D. Song. BACKDOORL: Backdoor Attack against Competitive Reinforcement Learning. IJCAI, 2021.
- [4] J. Cui, Y. Han, Y. Ma, J. Jiao, and J. Zhang. BadRL: Sparse Targeted Backdoor Attack Against Reinforcement Learning. AAAI, 2024.
- [5] H. Birkholz, D. Thaler, M. Richardson, N. Smith, and W. Pan. RFC 9334: Remote ATestation procedureS (RATS) Architecture. IETF, 2023.